Derivative Norm Transfer for the CLMS Jacobian Problem

Candidate partial result packet for arXiv:2208.13576

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Source Question

The source paper is Sauli Lindberg, *A note on the Jacobian problem of Coifman, Lions, Meyer and Semmes*, arXiv:2208.13576. In Corollary 1.11 the paper proves that, for the planar Jacobian quadratic operator

$$Q\omega = |\mathcal{S}\omega|^2 - |\omega|^2 : L^2(\mathbb{C},\mathbb{C}) \to \mathcal{H}^1(\mathbb{C}),$$

the associated dual unit ball is not strictly convex:

$$\operatorname{ext}(\mathbb{B}_{X_Q^*}) \subsetneq \mathbb{S}_{X_Q^*}.$$

Immediately afterward, the author writes that it is unclear whether the analogous statement holds for the Gateaux derivative

$$(\omega,\gamma) \mapsto Q'_\omega\gamma : L^2(\mathbb{C},\mathbb{C})^2 \to \mathcal{H}^1(\mathbb{C}).$$

(vi) The norm and relative weak topologies also coincide in* $NA_{\|\cdot\|_{X_Q}}$.

The proof is presented in §6. The second main result concerns uniqueness of minimum norm solutions and is proved in §5.

Theorem 1.10. *Suppose Assumptions 1.2–1.3 and 1.8 hold, and let* $f \in \operatorname{ext}(\mathbb{S}_{X_Q^*})$. *Then the minimum norm solution* $\omega \in \mathscr{A}$ *of* $Q\omega = f$ *is unique up to multiplication by* $c \in \mathbb{S}_{\mathbb{K}}$.

In view of (1.6) it is natural to ask whether X_Q^* is strictly convex, that is, $\operatorname{ext}(\mathbb{B}_{X_Q^*}) = \mathbb{S}_{X_Q^*}$. In the case of the planar Jacobian, the answer is negative, as an immediate consequence of Theorems 1.9–1.10 and the non-uniqueness of general minimum norm solutions:

Corollary 1.11. *In the case of* $\omega \mapsto Q\omega = |\mathcal{S}\omega|^2 - |\omega|^2 : L^2(\mathbb{C},\mathbb{C}) \to \mathcal{H}^1(\mathbb{C})$ *we have* $\operatorname{ext}(\mathbb{B}_{X_Q^*}) \subsetneq \mathbb{S}_{X_Q^*}$.

It is unclear to the author whether an analogue of Corollary 1.11 holds for the Gâteaux derivative $(\omega,\gamma) \mapsto Q'_\omega\gamma \colon L^2(\mathbb{C},\mathbb{C})^2 \to \mathscr{H}^1(\mathbb{C})$.

The source also proves the block self-adjoint modification identity:

$$\widetilde{T}_b(f,g) = (T_b^* g, T_b f), \qquad \|\widetilde{T}_b\|_{H\oplus_2 H \to H\oplus_2 H} = \|T_b\|_{H\to H},$$

and identifies this modification with the derivative:

$$\langle \widetilde{T}_b(f,g),(f,g)\rangle = \langle b, Q'_f g\rangle.$$

1.2 holds but Assumption 1.3 does not, we use a natural self-adjoint modification of T_b. Its existence and uniqueness are guaranteed by the following simple lemma.

Lemma 2.9. *Suppose $A\colon H \to H$ is $\mathbb{K}$-linear. Then there exists a unique self-adjoint $\mathbb{K}$-linear operator $B\colon H \times H \to H \times H$ such that*

$$\langle B(x,y),(x,y)\rangle_{H\times H} = 2\mathbf{Re}\langle Ax,y\rangle_H \quad \text{for all } x,y \in H.$$

The operator B is of the form $B(x,y) = (A^*y, Ax)$ and satisfies $\|B\|_{H\times H\to H\times H} = \|A\|_{H\to H}$.

Proposition 2.10. *If a bilinear mapping $T\colon X^{**} \times H \to H$ satisfies Assumption 1.2, then the modified operator*

$$(b,f,g) \mapsto \widetilde{T}_b(f,g)\colon X^{**} \times (H\times H) \to H \times H, \qquad \widetilde{T}_b(f,g) := (T_b^* g, T_b f)$$

satisfies Assumptions 1.2–1.3.

If T satisfies Assumptions 1.2–1.3, then $\tilde{T}$ is related to the Gâteaux derivative of Q by

$$\langle \tilde{T}_b(f,g),(f,g)\rangle_{H\times H} = \langle b, Q'_f g\rangle_{X^{**}-X^{*}} \qquad \text{for all } f,g\in H,\ b\in X^{**}. \tag{2.2}$$

Claimed Partial Result

Theorem 1. *Let $Q : H \to X^*$ be a quadratic operator in the source paper's framework, generated by $T_b : H \to H$, and let $\widetilde{Q} : H \oplus_2 H \to X^*$ be the self-adjoint block modification*

$$\langle b, \widetilde{Q}(f,g)\rangle = \langle \widetilde{T}_b(f,g),(f,g)\rangle, \qquad \widetilde{T}_b(f,g) = (T_b^* g, T_b f).$$

Then the coefficient-space norm induced by $\widetilde{Q}$ is identical to the coefficient-space norm induced by Q:

$$\|b\|_{X_{\widetilde{Q}}} = \|b\|_{X_Q} \qquad (b \in X).$$

Consequently, $X_{\widetilde{Q}} = X_Q$ isometrically by the identity map, and the dual unit balls $\mathbb{B}_{X^*_{\widetilde{Q}}}$ and $\mathbb{B}_{X^*_Q}$ have the same extreme points.

Proof

By definition in the source framework,

$$\|b\|_{X_Q} = \sup_{\|\omega\|_H=1} \langle b, Q\omega\rangle = \|T_b\|_{H\to H}.$$

For the modified operator, the source's self-adjoint block construction gives

$$\langle b, \widetilde{Q}(f,g)\rangle = \langle \widetilde{T}_b(f,g),(f,g)\rangle$$

on $H \oplus_2 H$, and $\widetilde{T}_b$ is self-adjoint. Therefore the induced norm is its numerical radius, which equals its operator norm:

$$\|b\|_{X_{\widetilde{Q}}} = \sup_{\|(f,g)\|_{H\oplus_2 H}=1} \langle \widetilde{T}_b(f,g),(f,g)\rangle = \|\widetilde{T}_b\|_{H\oplus_2 H\to H\oplus_2 H}.$$

The source lemma gives

$$\|\widetilde{T}_b\|_{H\oplus_2 H\to H\oplus_2 H} = \|T_b\|_{H\to H}.$$

Combining the three displayed identities yields

$$\|b\|_{X_{\widetilde{Q}}} = \|T_b\| = \|b\|_{X_Q}.$$

Thus the identity map is an isometry from X_Q onto $X_{\widetilde{Q}}$. The dual identity is also an isometry, so the dual unit balls are literally the same convex body in X^*. Their extreme points and strict convexity properties are therefore identical.

Derivative Consequence

Corollary 1. *For the planar Jacobian in arXiv:2208.13576, let $\widetilde{Q}$ denote the Gateaux derivative operator*

$$\widetilde{Q}(\omega, \gamma) = Q'_\omega \gamma : L^2(\mathbb{C}, \mathbb{C})^2 \to \mathcal{H}^1(\mathbb{C}).$$

Then

$$\mathrm{ext}(\mathbb{B}_{X^*_{\widetilde{Q}}}) \subsetneq \mathbb{S}_{X^*_{\widetilde{Q}}}.$$

In particular, the analogue of Corollary 1.11 for the Gateaux derivative holds.

Proof. The theorem identifies $X_{\widetilde{Q}}$ isometrically with X_Q. Hence

$$\mathrm{ext}(\mathbb{B}_{X^*_{\widetilde{Q}}}) = \mathrm{ext}(\mathbb{B}_{X^*_Q}), \qquad \mathbb{S}_{X^*_{\widetilde{Q}}} = \mathbb{S}_{X^*_Q}.$$

The source paper's Corollary 1.11 gives $\mathrm{ext}(\mathbb{B}_{X^*_Q}) \subsetneq \mathbb{S}_{X^*_Q}$ for the planar Jacobian. The same proper inclusion follows for $X^*_{\widetilde{Q}}$. □

Limitations and Novelty Check

This answers only the strict-convexity geometry remark for the Gateaux derivative. It does not prove $Q(\mathcal{A}) = \mathbb{S}_{X^*_Q}$, does not solve Question 1.6, and does not settle the CLMS/Iwaniec Jacobian surjectivity problem.

Cheap duplicate checks were run against the run registry, solution, attempt, and proof-gap indexes for arXiv:2208.13576 and for the core phrases "Gateaux derivative", "strictly convex", "self-adjoint modification", and "X_Q". No existing packet for this observation was found. Web exact-phrase checks found the source paper but no direct later answer to this derivative strict-convexity remark.

References

1. S. Lindberg, *A note on the Jacobian problem of Coifman, Lions, Meyer and Semmes*, arXiv:2208.13576.

2. R. Coifman, P.-L. Lions, Y. Meyer, and S. Semmes, *Compensated compactness and Hardy spaces*, J. Math. Pures Appl. 72 (1993), 247–286.

Human Review Recommendation

Review as a likely valid partial answer to the source remark. The main point to verify is not analytical but notational: the derivative operator in the remark should be read with the Hilbert direct-sum norm used in the source's self-adjoint modification. Under that convention, the induced X_Q-norms are identical.